Do Multivariate Gene Expression Patterns Predict Donor Age Within and Across Human Tissues?

C&S BIO 100 Final Project

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Biological Motivation

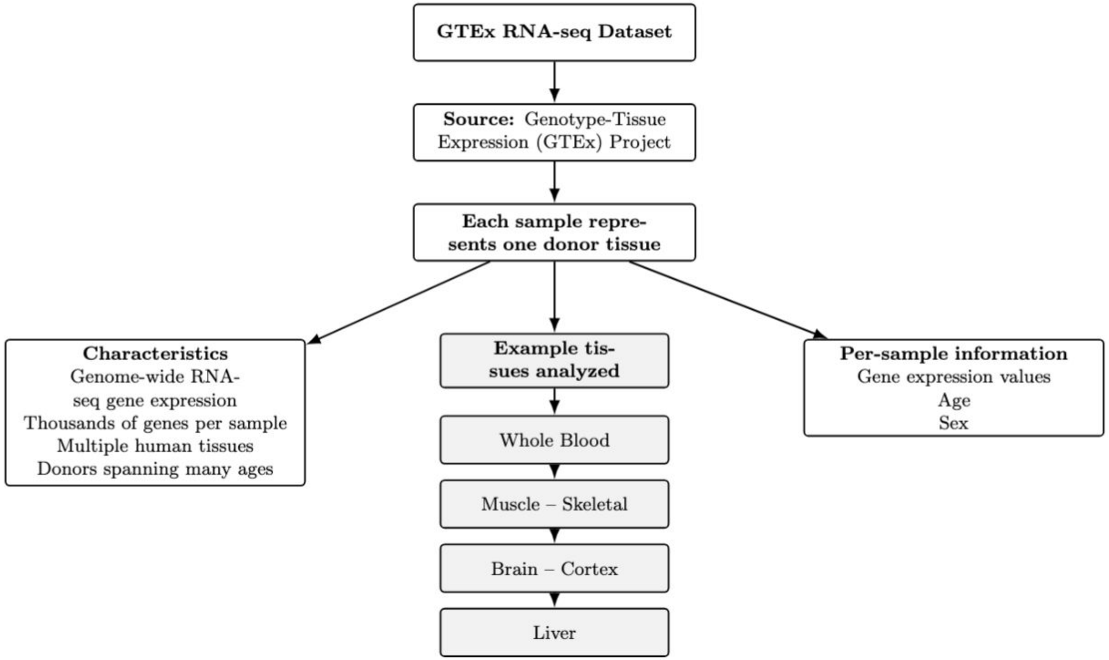
- Why study aging and gene expression?
  - Aging alters gene regulation and cellular function
  - These changes affect many genes at the same time
  - Patterns in gene expression across multiple genes may indicate aging
  - Understanding these patterns may help reveal how aging differs across tissues

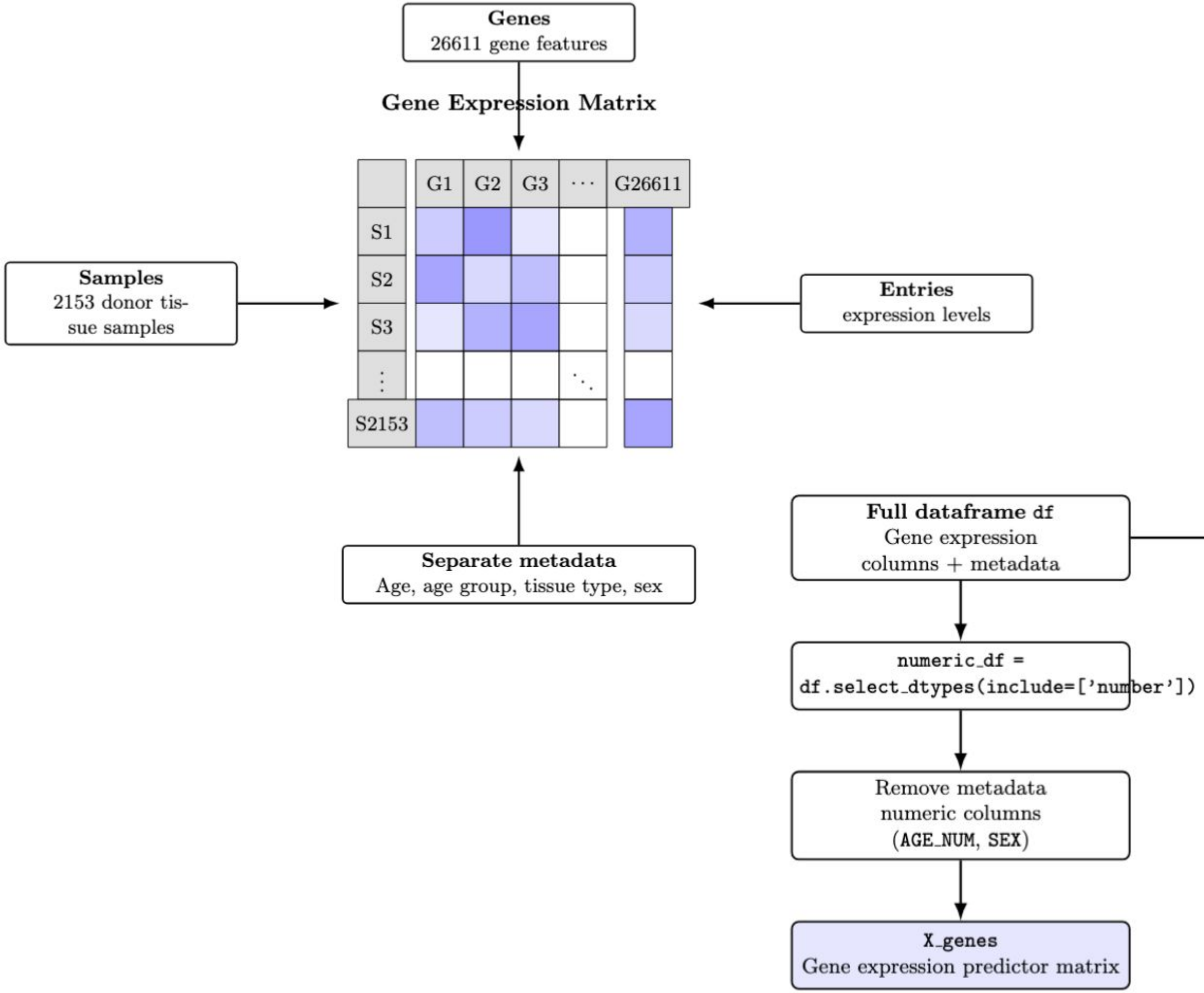
Research Question & Hypothesis

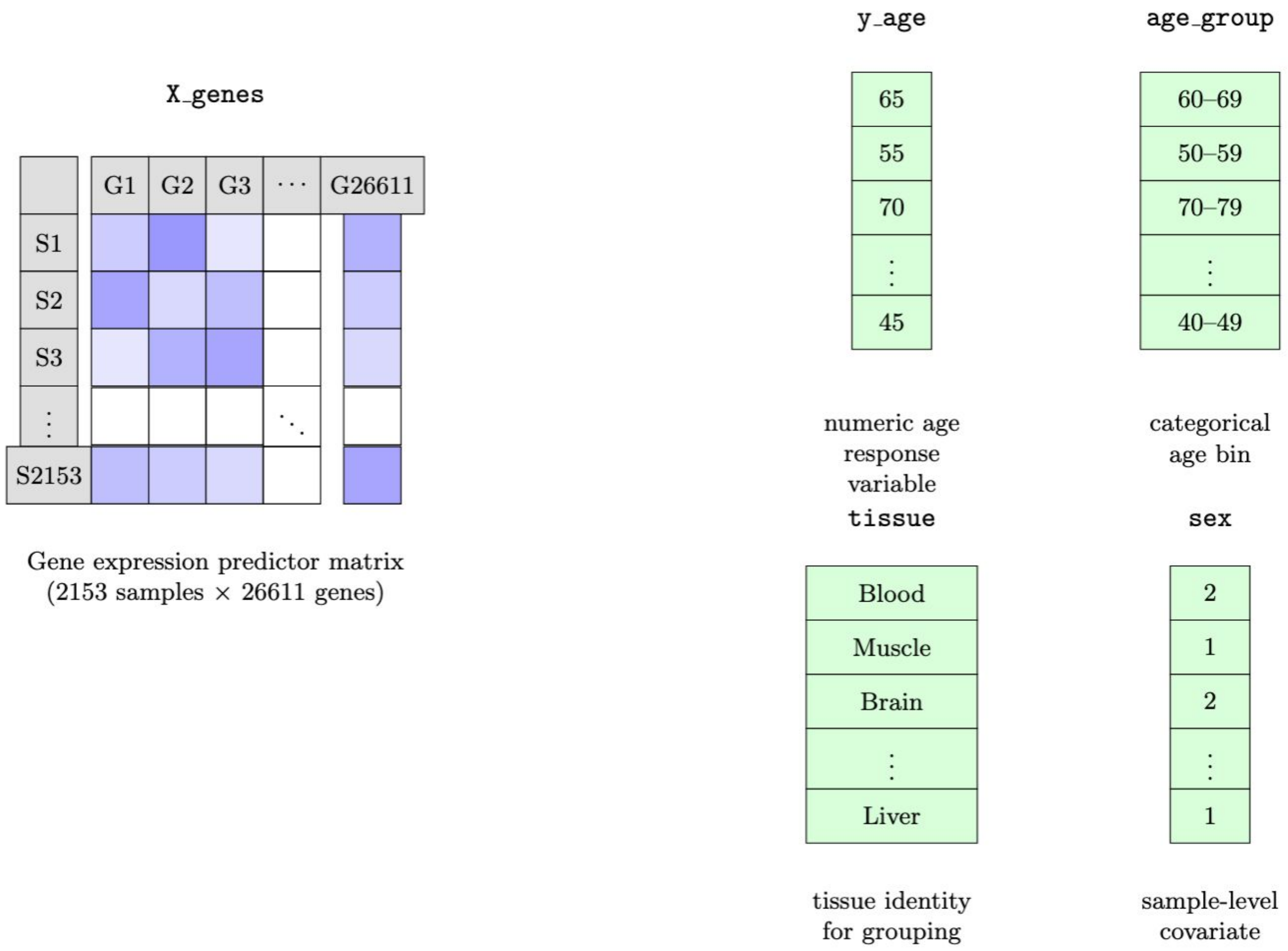
- Research Question:
  - Can multivariate gene expression patterns predict donor age within and across human tissues?
- Hypothesis
  - Gene expression patterns exhibit age-related structures
  - Some aging signals are shared across tissues
  - Some aging signals are tissue-specific

Dataset

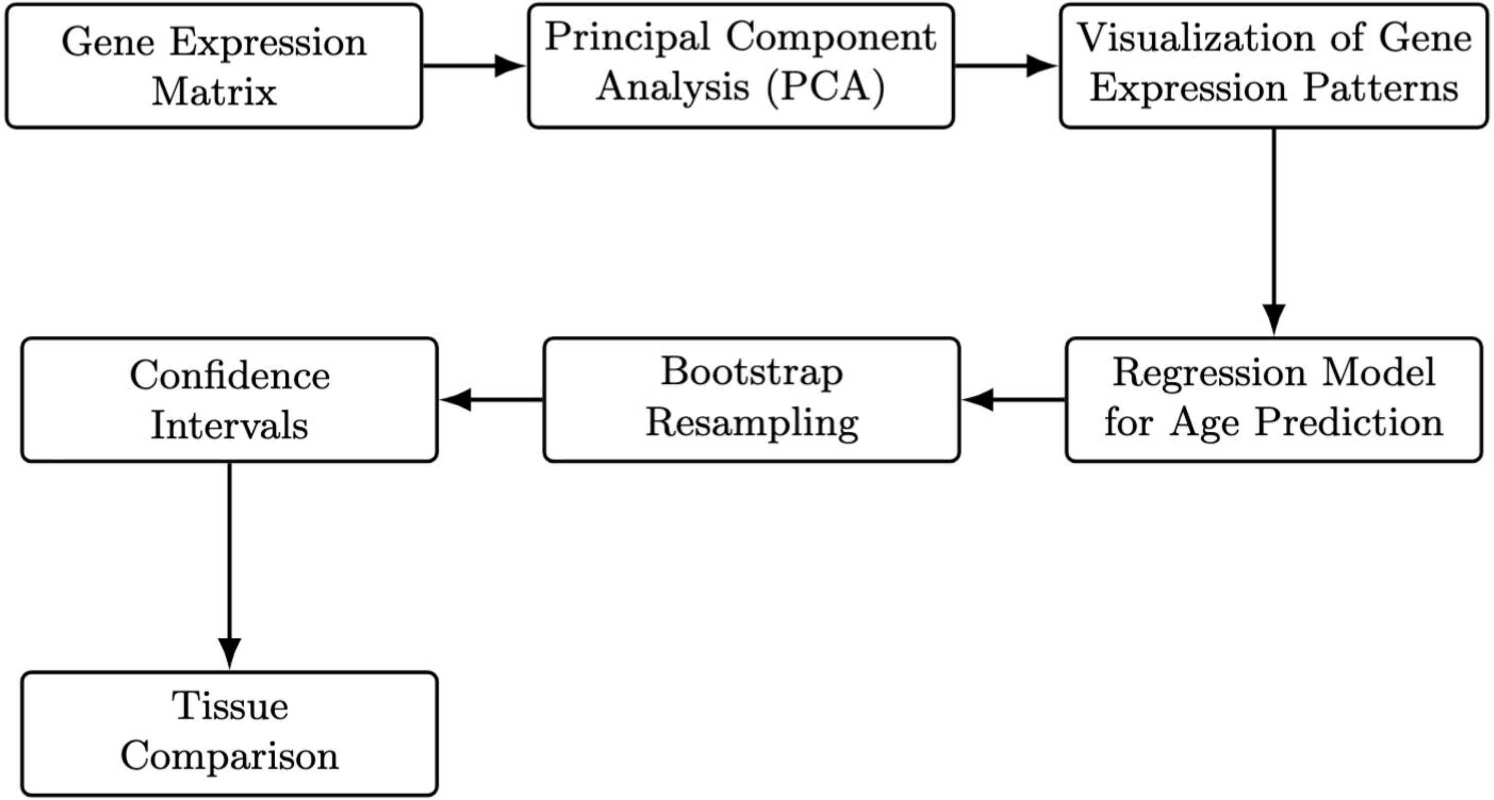
- GTEx RNA-seq Dataset
- Source:
    - Genotype Tissue Expression (GTEx) Project
- Characteristics:
    - Genome-wide RNA-seq gene expression
    - Thousands of genes per sample
    - Multiple human tissues
    - Donors spanning many ages
- Example tissues analyzed:
    - Whole Blood
    - Muscle - Skeletal
    - Brain - Cortex
    - Liver
- Each sample represents one donor tissue with expression values for numerous genes and also their age and gender




Genes
26611 gene features
Gene Expression Matrix
G1
G2
G3
· · ·
G26611
S1
S2
S3
S2153
Samples
2153 donor tissue samples
Entries
expression levels
Separate metadata
Age, age group, tissue type, sex
Full dataframe df
Gene expression columns + metadata
Metadata extracted from df
numeric_df = df.select_dtypes(include=['number'])
Remove metadata numeric columns (AGE_NUM, SEX)
X_genes
Gene expression predictor matrix
y_age
numeric age
age_group
tissue
sex


X_genes
G1
G2
G3
···
G26611
S1
S2
S3
⋮
S2153
Gene expression predictor matrix
(2153 samples × 26611 genes)
y_age
65
55
70
⋮
45
numeric age
response
variable
age_group
60–69
50–59
70–79
⋮
40–49
categorical
age bin
tissue
Blood
Muscle
Brain
⋮
Liver
tissue identity
for grouping
sex
2
1
2
⋮
1
sample-level
covariate

Analysis Workflow

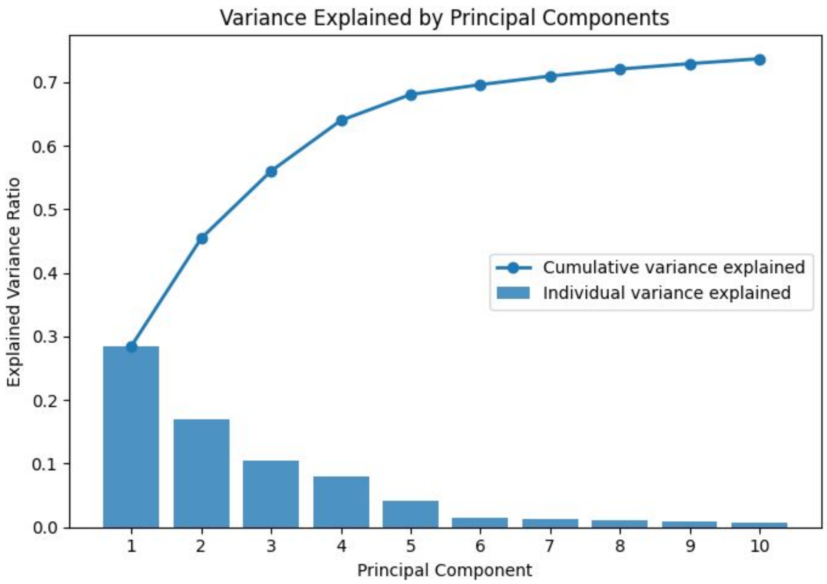


Unsupervised Analysis: PCA

- Purpose
  - Reduce dimensionality of gene expression data
  - Identify dominant variation patterns
- Key Properties
  - PCA finds directions of maximum variance
  - Each principal component is a weighted combination of genes
  - PCAs do not use age information
- If age structure appears in the PCA, it comes out naturally from the gene expressions

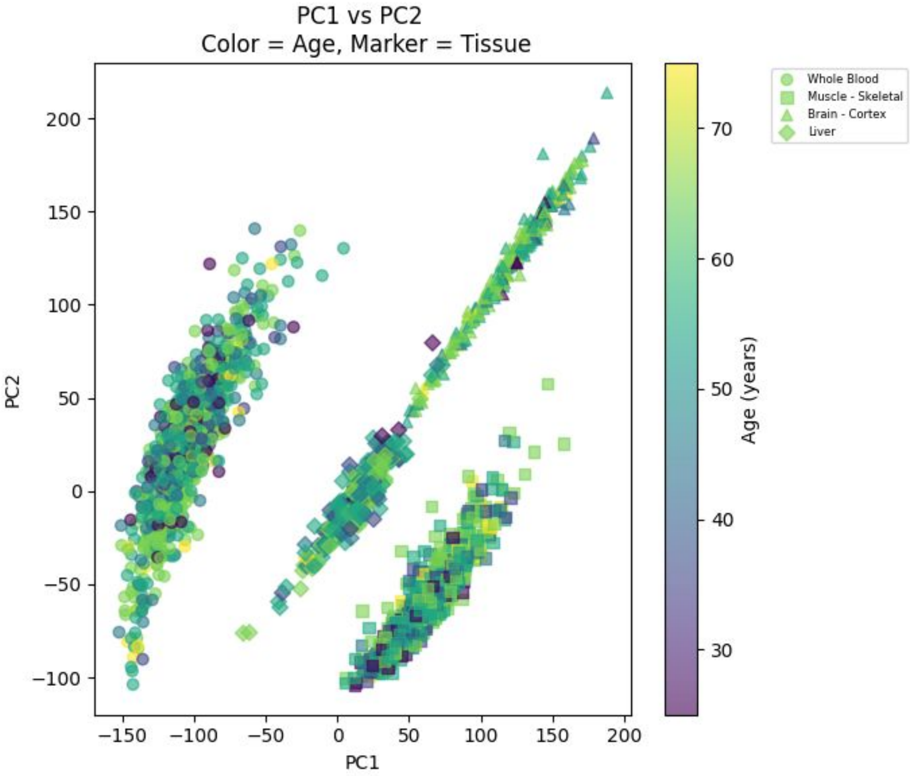
Variance Explained by Principal Components

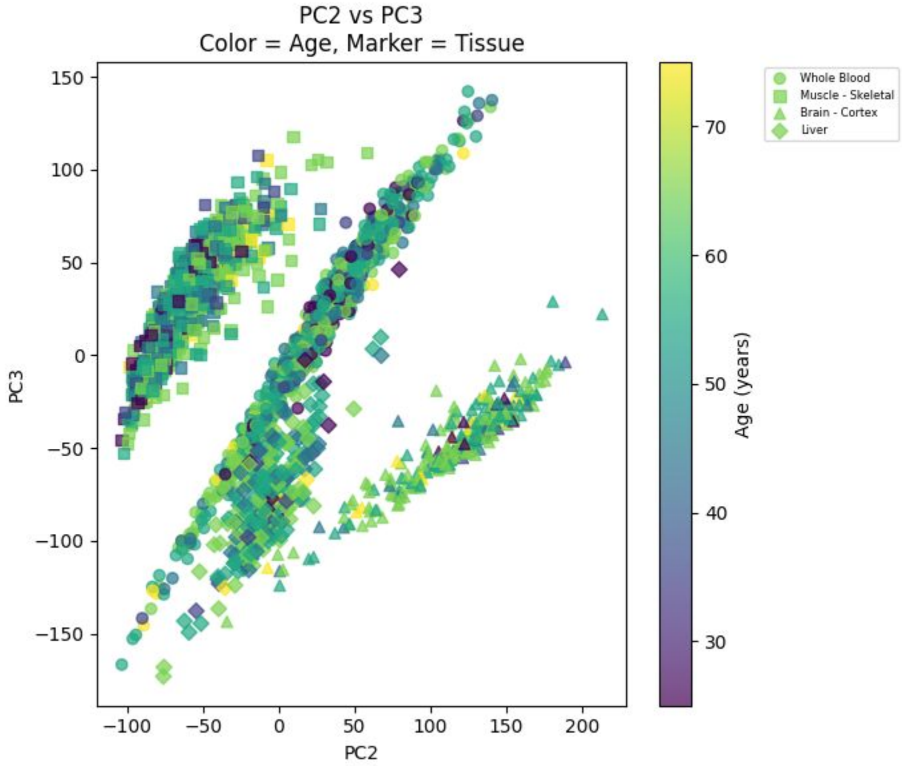
- PC1 explains 28.52% of variance in gene expression
- PC1 - PC3 explains 56.01% of variance
- PC1 - PC10 explains 73.67% of variance
- Later components contribute progressively less information



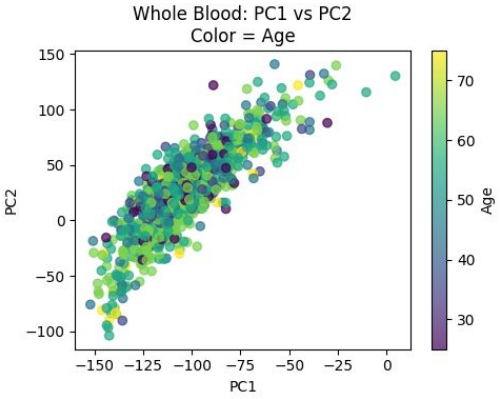
PC1: 28.52% | cumulative 28.52%
PC2: 16.93% | cumulative 45.45%
PC3: 10.56% | cumulative 56.01%
PC4: 7.96% | cumulative 63.97%
PC5: 4.08% | cumulative 68.05%
PC6: 1.55% | cumulative 69.61%
PC7: 1.35% | cumulative 70.96%
PC8: 1.10% | cumulative 72.05%
PC9: 0.85% | cumulative 72.91%
PC10: 0.77% | cumulative 73.68%

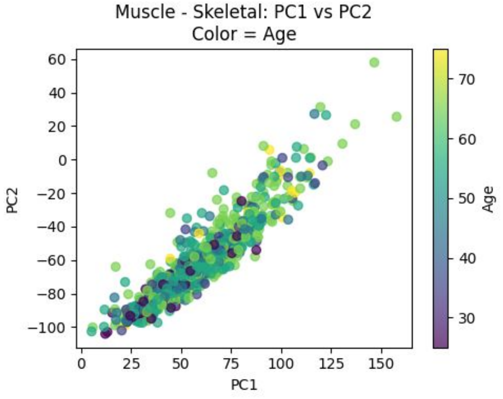
PCA Results (All Tissues)

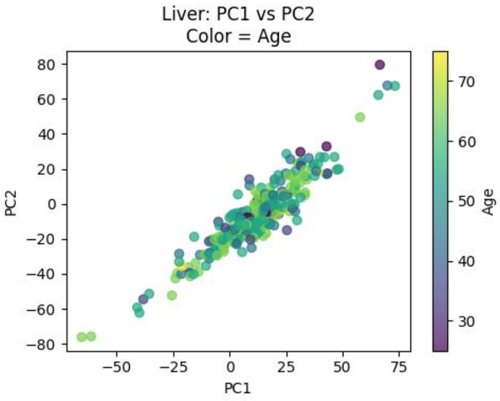


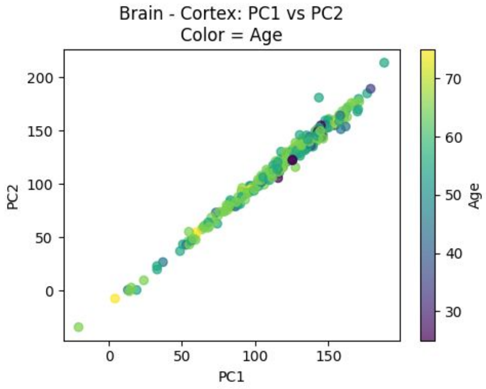


PCA within Tissues

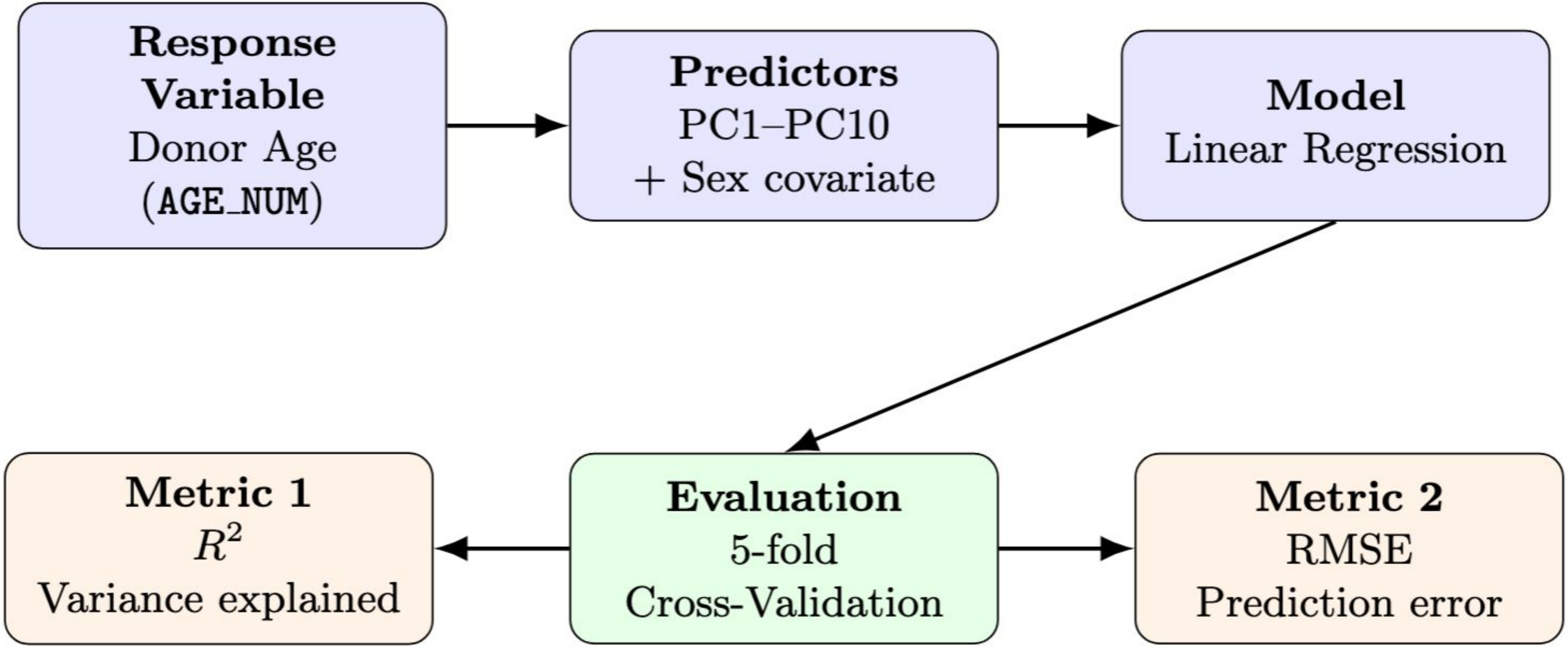


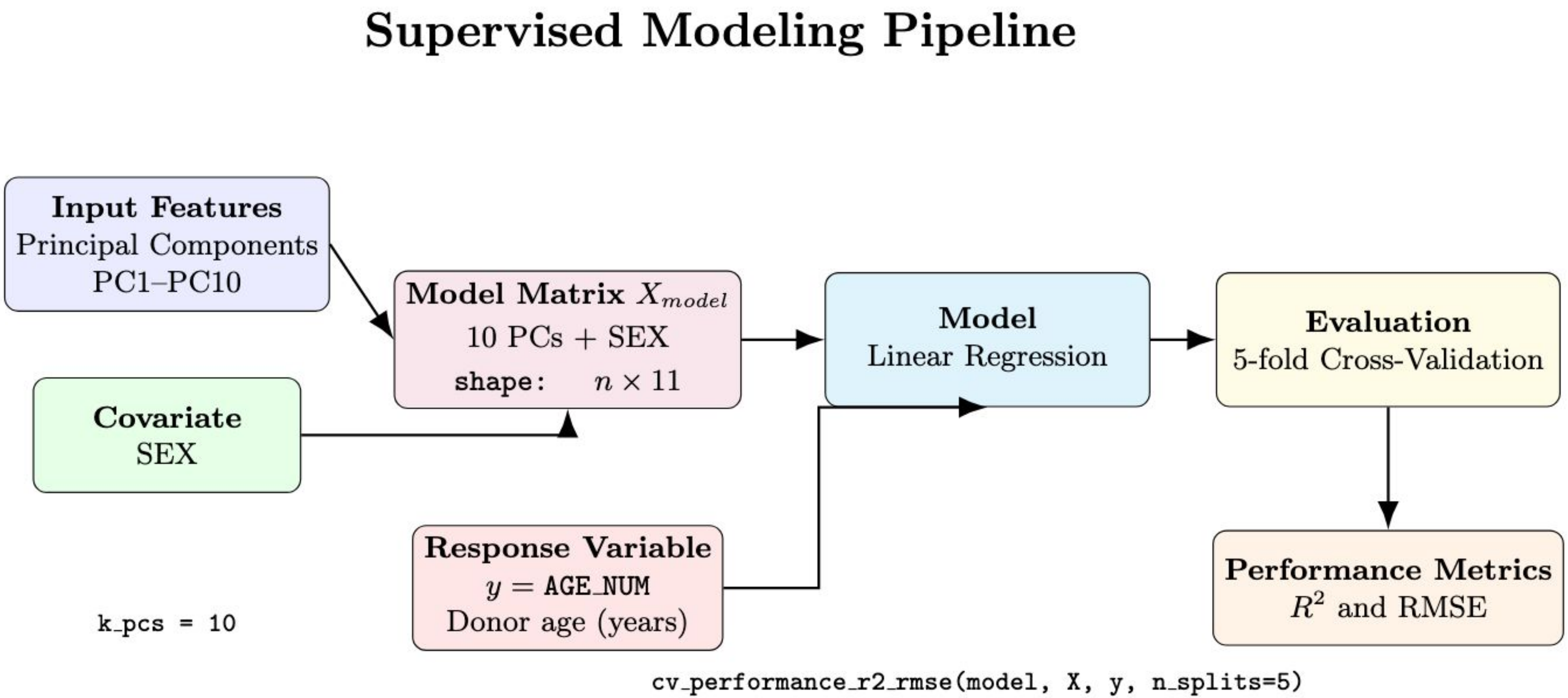






Supervised Modeling




Supervised Modeling Pipeline
Input Features
Principal Components
PC1–PC10
Model Matrix X_{model}
10 PCs + SEX
shape: $n \times 11$
Model
Linear Regression
Evaluation
5-fold Cross-Validation
Covariate
SEX
Response Variable
y = AGE_NUM
Donor age (years)
Performance Metrics
R^2 and RMSE
k_pcs = 10
cv_performance_r2_rmse(model, X, y, n_splits=5)

Model Performance

Cross-validated linear regression using predictors PC1–PC10 + Sex

R^2

0.129

Explains ≈13% of age variation

RMSE

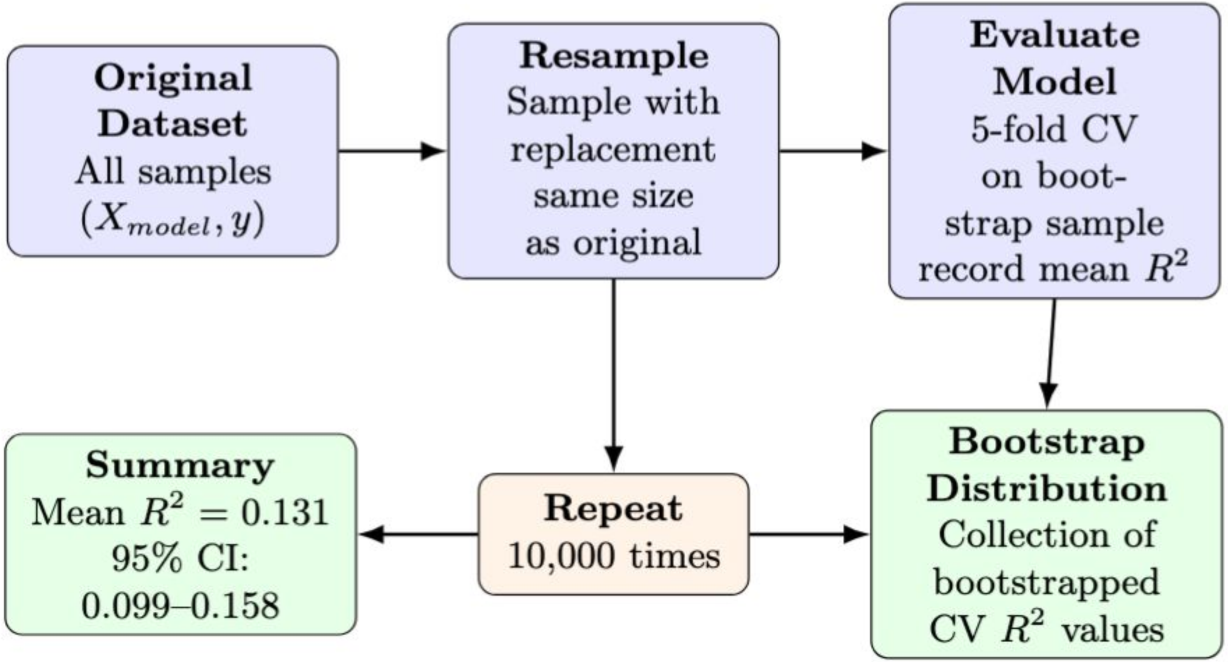
11.7 years

Average prediction error

Interpretation:

Gene expression patterns summarized by the first 10 principal components, along with sex, explain a modest portion of donor age variability. Predictions differ from the true age by about 12 years on average.

How the Bootstrap Resampling Works



Bootstrap Analysis

Bootstrap resampling to assess uncertainty and stability of cross-validated model performance

Resampling

10,000 bootstrap iterations

Mean CV R^2

0.130
across bootstrap samples

95% CI

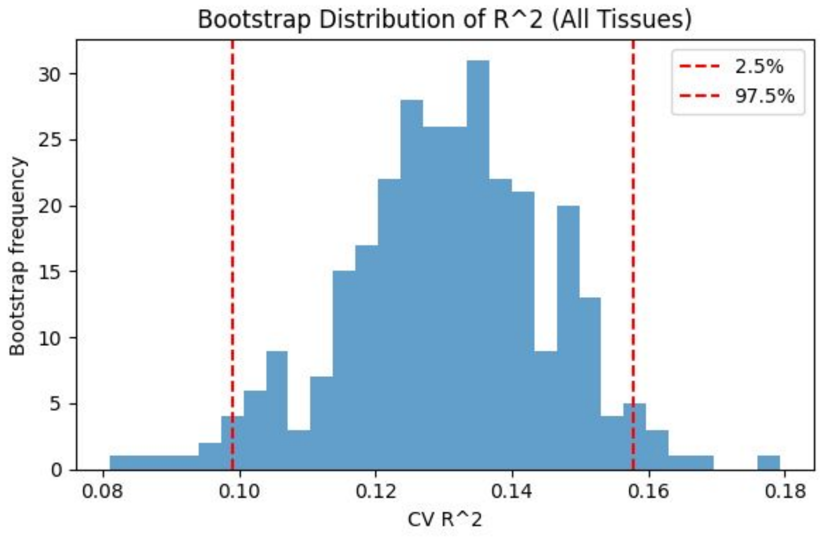
0.102 – 0.160
for bootstrapped CV R^2

Interpretation:

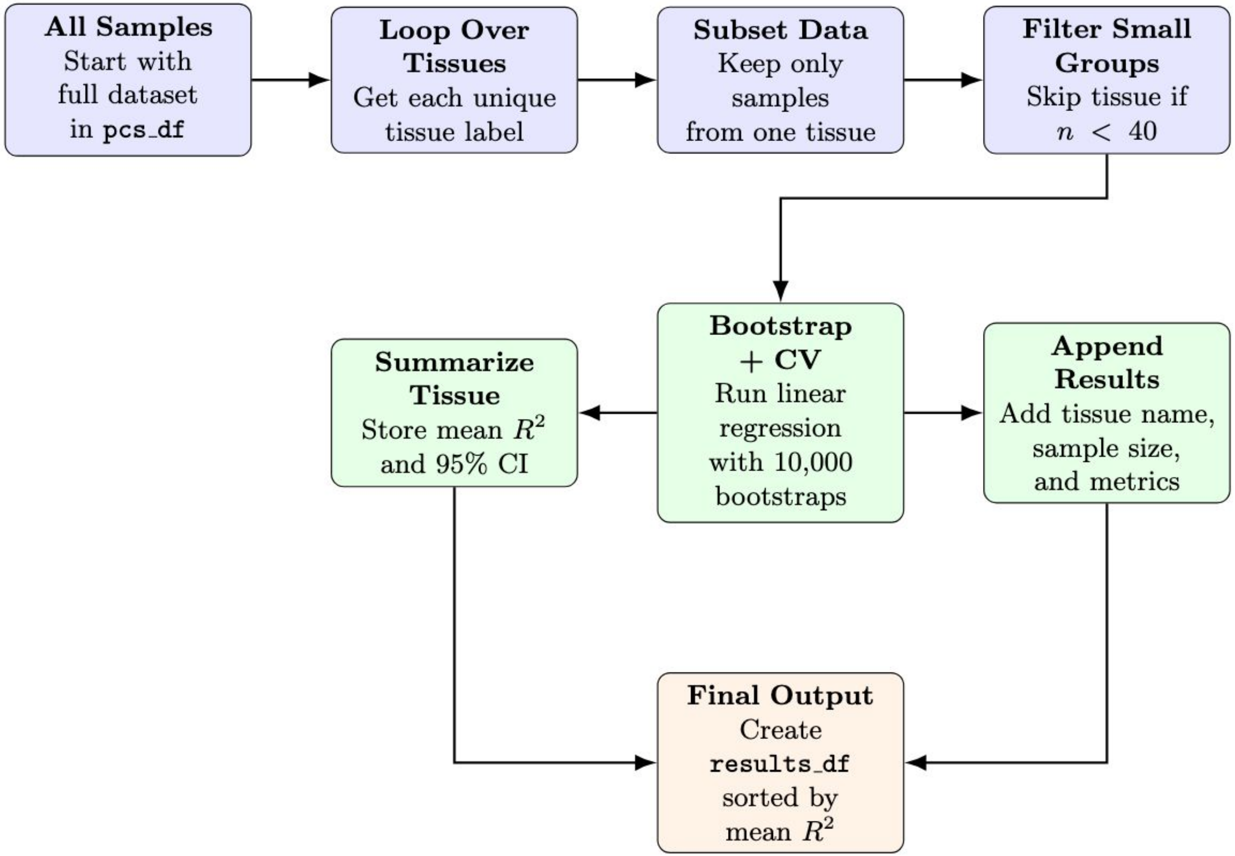
Bootstrap resampling shows that model performance is fairly stable across repeated samples. The average cross-validated R^2 is about 0.130, and the 95% confidence interval from 0.102 to 0.160 suggests modest but consistently positive predictive signal.

Bootstrap Distribution

- Distribution centered around R^2 ≈ 0.13
- Red dashed lines show the 95% CI
- Performance remains consistently above zero



Tissue-Specific Bootstrap Analysis Pipeline



Tissue-Specific Bootstrap Results

Whole Blood
Sample size: $n = 803$

Mean R^2: 0.170
95% CI: [0.118, 0.223]

Muscle - Skeletal
Sample size: $n = 818$

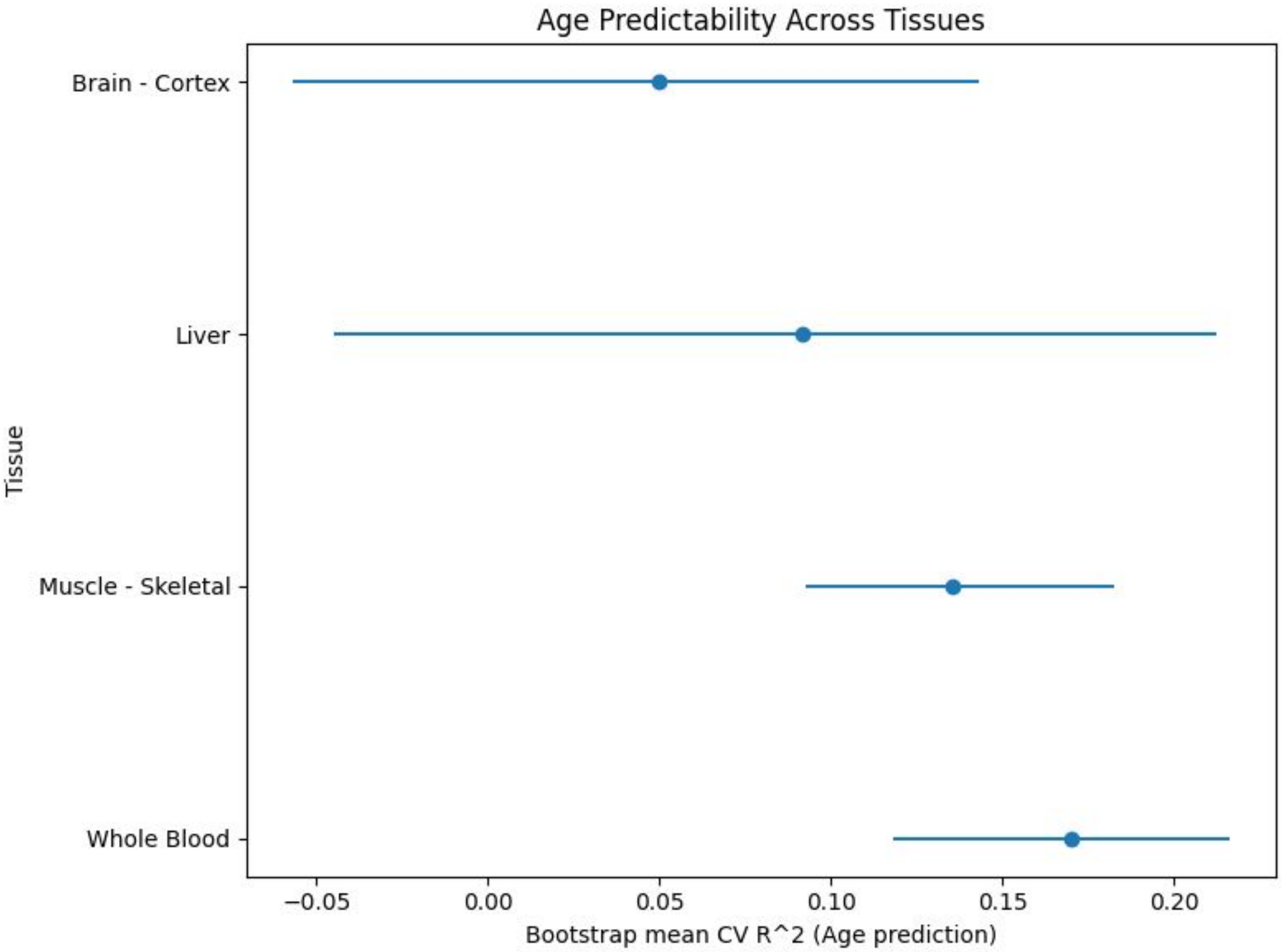
Mean R^2: 0.139
95% CI: [0.092, 0.186]

Liver
Sample size: $n = 262$

Mean R^2: 0.091
95% CI: [-0.027, 0.212]

Brain - Cortex
Sample size: $n = 270$

Mean R^2: 0.049
95% CI: [-0.067, 0.157]


Age Predictability Across Tissues
Brain - Cortex
Liver
Muscle - Skeletal
Whole Blood
Tissue
−0.05
0.00
0.05
0.10
0.15
0.20
Bootstrap mean CV R^2 (Age prediction)

Conclusion and Interpretation

- Key Findings
  - PCA shows that tissue identity dominates among gene expression variation
  - Gene expression patterns contain a modest aging signal
  - Linear regression using PC1 - PC10 + sex explains about 13% of age variation
  - Age predictability differs across tissue
- Biological Interpretation
  - Whole blood shows the strongest age signal
  - Muscle shows moderate age predictability
  - Liver and brain cortex show weaker signal
  - Suggesting that aging-related transcriptomic changes may be tissue-specific, potentially reflecting differences in cellular turnover and biological function across tissues

Pitfalls and Future Directions

- Limitations
  - PCA captures only linear variation
  - Aging biology may involve nonlinear gene interactions
  - GTEx data is cross-sectional and not vertical
  - Environmental and lifestyle variables not included
- Future Directions
  - Test nonlinear models to capture these aging patterns
  - Explore regularized models like Elastic Net (done in next slide)
    - Combine LASSO + Ridge penalties to reduce overfitting

Bonus Analysis: Elastic Net